

# Inyoung Ryu

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## RESEARCH INTERESTS

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- Surface group representations and character varieties
- Mapping class groups and Teichmüller theory
- Low-dimensional topology and geometric group theory

## EDUCATION

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| <b>Texas A&amp;M University</b><br><i>Ph.D. in Mathematics. Expected graduation date: May 2027</i>                                                                                                                   | Aug. 2021 – Present<br>Texas, US          |
| <b>Ulsan National Institute of Science and Technology (UNIST)</b><br><i>Bachelor of Science in Mathematical Sciences</i> <ul style="list-style-type: none"><li>• Minor in Computer Science and Engineering</li></ul> | Mar. 2017 – Jul. 2021<br>Ulsan, Korea     |
| <b>Sabancı University</b><br><i>Erasmus Program</i>                                                                                                                                                                  | Feb. 2020 – Jun. 2020<br>Istanbul, Turkey |

## PUBLICATIONS AND PREPRINTS

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| <b>From Bowditch's question to Goldman's conjecture for type-preserving representations</b><br><i>with Viraj Joshi, submitted, arXiv: 2609.02065</i> | Sep. 2026 |
| <b>On totally hyperbolic non-Fuchsian type-preserving representations</b><br><i>Submitted, arXiv: 2511.13989</i>                                     | Nov. 2025 |
| <b>Connected components of the space of type-preserving representations</b><br><i>with Tian Yang, submitted, arXiv: 2507.07391</i>                   | Jul. 2025 |

## INVITED SEMINARS AND COLLOQUIUM TALKS

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| <b>(Upcoming) Two invariants of type-preserving representations and applications</b><br><i>Joint Mathematics Meetings (JMM 2027)</i>                                  | Jan. 2027<br>Illinois, US    |
| <b>(Upcoming) Two invariants of type-preserving representations and applications</b><br><i>Geometric Group Theory Seminar, Purdue University</i>                      | Nov. 2026<br>Indiana, US     |
| <b>(Upcoming) Two invariants of type-preserving representations and applications</b><br><i>RTG Seminar on Geometry, Dynamics and Topology, University of Michigan</i> | Nov. 2026<br>Michigan, US    |
| <b>Two invariants of type-preserving representations and applications</b><br><i>Geometry and Dynamics Seminar, Centro de Ciencias Matemáticas UNAM (Online)</i>       | Sep. 2026<br>Morelia, Mexico |
| <b>Two invariants of type-preserving representations and applications</b><br><i>Topology Seminar, University of Tennessee</i>                                         | Apr. 2026<br>Tennessee, US   |
| <b>Two invariants of type-preserving representations and applications</b><br><i>Topology &amp; Geometry Seminar, Rutgers University</i>                               | Apr. 2026<br>New Jersey, US  |
| <b>Two invariants of type-preserving representations and applications</b><br><i>Algebra &amp; Combinatorics Seminar, Texas A&amp;M University</i>                     | Mar. 2026<br>Texas, US       |
| <b>Two invariants of type-preserving representations and applications</b><br><i>Groups &amp; Dynamics Seminar, University of Texas Austin</i>                         | Mar. 2026<br>Texas, US       |
| <b>Two invariants of type-preserving representations and applications</b><br><i>Groups &amp; Dynamics Seminar, Texas A&amp;M University</i>                           | Feb. 2026<br>Texas, US       |
| <b>Connected components of the space of type-preserving representations</b><br><i>SHSU Junior Colloquium, Sam Houston State University</i>                            | Nov. 2025<br>Texas, US       |
| <b>Connected components of the space of type-preserving representations</b><br><i>KIAS HCMC Topology Seminar, Korea Institute for Advanced Study (Online)</i>         | Feb. 2025<br>Seoul, Korea    |

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| <b>Geometric approach to surface group representations</b><br><i>Graduate Student Organization Seminar, Texas A&amp;M University</i> | Sep. 2024<br>Texas, US    |
| <b>Topological components of spaces of representations</b><br><i>SNU Topology Seminar, Seoul National University (Online)</i>        | Aug. 2024<br>Seoul, Korea |
| <b>Mapping class group action on Teichmüller Space</b><br><i>Graduate Student Organization Seminar, Texas A&amp;M University</i>     | Apr. 2024<br>Texas, US    |

## CONTRIBUTED TALKS

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| <b>(Upcoming) On totally hyperbolic non-Fuchsian type-preserving representations</b><br><i>AMS 2026 Fall Western Sectional Meeting, Arizona State University</i>           | Nov. 2026<br>Arizona, US       |
| <b>(Upcoming) Non-Fuchsian totally hyperbolic type-preserving representations</b><br><i>AMS 2026 Fall Central Sectional Meeting, University of Minnesota Duluth</i>        | Oct. 2026<br>Minnesota, US     |
| <b>Connected components of the space of type-preserving representations</b><br><i>GTA: Philadelphia 2025, Temple University</i>                                            | May 2025<br>Pennsylvania, US   |
| <b>Connected components of the space of type-preserving representations</b><br><i>CIRM School on Geometric structures and discrete group actions</i>                       | Apr. 2025<br>Marseille, France |
| <b>Connected components of the space of type-preserving representations</b><br><i>Graduate Student Conference in Topology and Geometry, Indiana University Bloomington</i> | Apr. 2025<br>Indiana, US       |
| <b>Nielsen-Thurston classification</b><br><i>GIGEM Conference, Texas A&amp;M University</i>                                                                                | Apr. 2024<br>Texas, US         |

## LIGHTNING TALKS AND POSTER PRESENTATIONS

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| <b>Can a non-Fuchsian representation be totally hyperbolic?</b><br><i>Vanderbilt colloquium series: Groups, geometry, and dynamics (ICM satellite conference)</i> | Jul. 2026<br>Tennessee, US    |
| <b>Poster Presentation: Non-Fuchsian totally hyperbolic representations</b><br><i>Young Geometric Group Theory XIV</i>                                            | Jun. 2026<br>Seoul, Korea     |
| <b>Connected components of the space of type-preserving representations</b><br><i>Ventotene Workshop on Higher-dimensional hyperbolic geometry</i>                | Sep. 2025<br>Ventotene, Italy |
| <b>Connected components of the space of type-preserving representations</b><br><i>2K-GATE Workshop, Korea Institute for Advanced Study</i>                        | Jun. 2025<br>Seoul, Korea     |
| <b>Connected components of the space of type-preserving representations</b><br><i>53rd Barrett Lectures: Recent Development in Geometric Group Theory</i>         | May 2025<br>Tennessee, US     |
| <b>Connected components of the space of type-preserving representations</b><br><i>Conference on Geometry and Dynamics in Low Dimensions, IMVAST</i>               | Jan. 2025<br>Hanoi, Vietnam   |

## SELECTED CONFERENCES AND WORKSHOPS

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| <b>International Congress of Mathematicians (ICM 2026)</b>                                                                 | Jul. 2026<br>Pennsylvania, US     |
| <b>Joint Mathematics Meetings (JMM 2026)</b>                                                                               | Jan. 2026<br>Washington, D.C., US |
| <b>Workshop on Homogeneous Dynamics and Anosov Representations</b><br><i>Simons Laufer Mathematical Sciences Institute</i> | Apr. 2026<br>California, US       |
| <b>SLMath Summer School: Geometry and Dynamics in Higher Rank Lie Groups</b><br><i>University of California, Berkeley</i>  | Jul. 2025<br>California, US       |
| <b>Workshop on Low-dimensional phenomena: geometry and dynamics</b><br><i>Institut Henri Poincaré</i>                      | Jun. 2025<br>Paris, France        |
| <b>Workshop on Lorentzian, Affine, and Hyperbolic Differential Geometry</b><br><i>University of Maryland</i>               | Apr. 2025<br>Maryland, US         |
| <b>Workshop on Motives and Mapping Class Groups</b><br><i>American Institute of Mathematics, Caltech</i>                   | Jan. 2025<br>California, US       |
| <b>Joint Mathematics Meetings (JMM 2024)</b>                                                                               | Jan. 2024<br>California, US       |
| <b>Conference on Characters and Moduli of Surfaces</b><br><i>Kyoto University</i>                                          | Jul. 2023<br>Kyoto, Japan         |

## HONORS, AWARDS, AND GRANTS

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| <b>AMS Fall 2026 Sectional Meeting Travel Grant</b><br><i>American Mathematical Society</i>                                                 | Sep. 2026<br>Texas, US             |
| <b>Graduate Research Assistant, supported by NSF Grant, PI: Tian Yang</b><br><i>Texas A&amp;M University</i>                                | Sep. 2026 - Dec. 2026<br>Texas, US |
| <b>Graduate Research Assistant, supported by NSF Grant, PI: Tian Yang</b><br><i>Texas A&amp;M University</i>                                | Jul. 2024 – Dec. 2024<br>Texas, US |
| <b>Houston A&amp;M Mothers' Club Outstanding TA Award</b><br><i>Awarded for going above and beyond expectations as a teaching assistant</i> | Jul. 2024<br>Texas, US             |
| <b>Graduate Research Assistant, supported by NSF Grant, PI: Tian Yang</b><br><i>Texas A&amp;M University</i>                                | Jan. 2024 – May 2024<br>Texas, US  |

## TEACHING

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| <b>Python Lab Assistant, MATH 151 &amp; 152 (Engineering Mathematics I &amp; II)</b><br><i>Texas A&amp;M University. Creating Python lab assignments.</i>              | Jul. 2026<br>Texas, US                |
| <b>Instructor of record, MATH 142 (Business Calculus)</b><br><i>Texas A&amp;M University. Lecturing, holding office hours. Writing, proctoring, and grading exams.</i> | Jan. 2026 - May 2026<br>Texas, US     |
| <b>Grader, MATH 636 (Topology I)</b><br><i>Texas A&amp;M University</i>                                                                                                | Sep. 2025 - Dec. 2025<br>Texas, US    |
| <b>Python Lab Assistant, MATH 150 (Functions, Trigonometry and Linear Systems)</b><br><i>Texas A&amp;M University. Creating Python lab assignments.</i>                | Jul. 2025<br>Texas, US                |
| <b>Grader, MATH 467 (Modern Geometry)</b><br><i>Texas A&amp;M University</i>                                                                                           | Jan. 2025 - May 2025<br>Texas, US     |
| <b>Help Session Tutor, MATH 142 (Business Calculus)</b><br><i>Texas A&amp;M University</i>                                                                             | Jun. 2024<br>Texas, US                |
| <b>Grader, MATH 636 (Topology I)</b><br><i>Texas A&amp;M University</i>                                                                                                | Aug. 2023 – Dec. 2023<br>Texas, US    |
| <b>Grader, MATH 142 (Business Calculus)</b><br><i>Texas A&amp;M University</i>                                                                                         | Jul. 2023<br>Texas, US                |
| <b>Python Lab Assistant, MATH 151 (Engineering Math I)</b><br><i>Texas A&amp;M University. Creating Python lab assignments.</i>                                        | Jun. 2023<br>Texas, US                |
| <b>Teaching Assistant, MATH 152 (Engineering Math II)</b><br><i>Texas A&amp;M University. Leading recitations.</i>                                                     | Jan. 2023 – May 2023<br>Texas, US     |
| <b>Teaching Assistant, MATH 151 (Engineering Math I)</b><br><i>Texas A&amp;M University. Leading recitations.</i>                                                      | Aug. 2022 – Dec. 2022<br>Texas, US    |
| <b>Grader, MATH 304 (Linear Algebra)</b><br><i>Texas A&amp;M University</i>                                                                                            | Jun. 2022<br>Texas, US                |
| <b>Grader, MATH 423 (Linear Algebra II)</b><br><i>Texas A&amp;M University</i>                                                                                         | Jan. 2022 – May 2022<br>Texas, US     |
| <b>Grader, MATH 467 (Modern Geometry)</b><br><i>Texas A&amp;M University</i>                                                                                           | Aug. 2021 – Dec. 2021<br>Texas, US    |
| <b>Teaching Assistant (Calculus I, Calculus II)</b><br><i>Ulsan National Institute of Science and Technology. Assistant in discussion sessions</i>                     | Mar. 2018 – Dec. 2019<br>Ulsan, Korea |
| <b>Calculus I Open Lecture</b><br><i>Ulsan National Institute of Science and Technology</i>                                                                            | Jun. 2018<br>Ulsan, Korea             |

## MENTORING

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| <b>Directed Reading Program: A first course in Algebraic Topology</b><br><i>Texas A&amp;M University</i> | Sep. 2025 - Dec. 2025<br>Texas, US |
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SERVICE

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**Treasurer, Association for Women in Mathematics**  
*Texas A&M University*

Aug. 2022 - May 2023  
*Texas, US*

- Women in Math Mentoring Lunch
- AWM Peer Mentoring Program

**Teaching Assistant for Math Circle**  
*Texas A&M Math Circle*

Nov. 2023  
*Texas, US*

**Grader for High School Math Contest**  
*Texas A&M High School Mathematics Contest*

Nov. 2023  
*Texas, US*

**Grader for High School Math Contest**  
*Texas A&M High School Mathematics Contest*

Nov. 2022  
*Texas, US*